

VeriTrust AI

Privacy-First Adaptive Identity Trust Framework
for Banking

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Team Name: AI Nexus

Team Members: Somesh Kumar Singh(Team
Leader)

Email:someshsingh2206@gmail.com

*Continuous Identity Validation for Secure Digital
Banking*



Problem Statement

Challenges in Digital Banking: Current Threat Landscape

- Account Takeover (ATO) through stolen credentials
- Fake KYC and onboarding fraud
- Unauthorized account recovery attempts
- Insider misuse of customer data
- New device and suspicious login risks
- Behavioral anomalies indicating compromised accounts

Existing Limitation

- Authentication occurs only at login
- Limited continuous monitoring
- Excessive OTP usage causes poor user experience



Our Solution – VeriTrust AI

Privacy-First Adaptive Identity Trust Framework

VeriTrust AI continuously evaluates the trustworthiness of customers, employees, and devices across banking channels using real-time risk assessment.

Key Solution Highlights:

- Continuous identity verification instead of one-time authentication.
- Dynamic Trust Score generated for every login, transaction, and account recovery request.
- AI-powered behavioral analytics to detect unusual user activities.
- Device fingerprinting to identify trusted and suspicious devices.
- Risk-based authentication that triggers OTP or additional verification only when necessary.
- Insider threat monitoring for detecting misuse of privileged access.
- Real-time fraud detection to prevent account takeover and KYC fraud.
- Privacy-first architecture with minimal data collection and secure processing.

System architecture

Frontend- Banking Portal , Security Dashboard

Backend Services-Risk Scoring Engine , Risk Scoring Engine
Device Trust Service , Behavioral Analytics Engine
Fraud Detection Service , Insider Threat Service

Infrastructure

Spring Boot APIs

MongoDB

Kafka

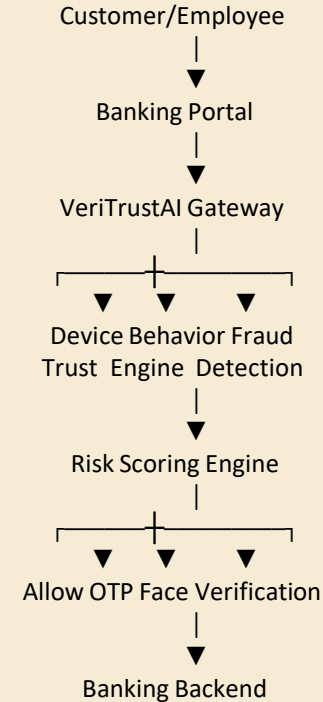
Python ML Services

Output

Trust Score

Risk Level

Verification Decision



Workflow

Step 1: User performs: Login , Transaction , Account Recovery

Step 2: System collects: Device Information, IP Address , Location, Browser Details, Behavioral Signals

Step 3:

Trust Score

Action

80-100

Allow

50-79

OTP

20-49

Face Verification

Below 20

Block

Key features



Device Trust

Trust scoring for endpoints, jailbreak/root detection, and provenance checks.

Behavioral Analytics

Continuous profiling: typing cadence, navigation patterns, transaction behavior.

Fraud Detection

Real-time anomaly detection and rule enrichment to catch KYC and payment fraud.

Insider Threat Detection

Baselines for privileged users, alerting on deviations and suspicious exports.

AI & data models

Datasets used to train and validate models — representative, anonymized where required, and augmented for rare events:

- PaySim dataset (simulated payments)
(<https://www.kaggle.com/datasets/ealaxi/paysim1>)
- Credit Card Fraud dataset (transaction fraud patterns)
(<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>)

- User Behavior & Login Detection Dataset

Purpose:

Behavioral anomaly detection, suspicious logins, and identity trust scoring.

(<https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset>)

Modeling strategy:



Isolation Forest

Unsupervised anomaly detection for novel behavior and device anomalies.



Random Forest

Robust classification for labeled fraud cases and explainability needs.



XGBoost

High-performance scoring for transaction risk with feature importance outputs.

Technology stack

Platform layers

Frontend	React.js, Tailwind CSS
Backend	Spring Boot, Spring Security, JWT
Database	MongoDB
AI/ML	Python, Scikit-Learn, XGBoost
Messaging	Apache Kafka
Deployment	Docker, Kubernetes, Render, Vercel

Designed for scalability, low latency, and secure orchestration of policy decisions.

Expected outcomes

Reduce account takeover incidents

Early detection of stolen credentials and session anomalies — fewer fraudulent logins.

Prevent KYC fraud

Layered identity signals increase onboarding accuracy and reduce synthetic accounts.

Detect insider threats

Behavioral baselines for staff limit lateral movement and unauthorized access.

Improve digital trust & UX

Adaptive friction keeps genuine customers moving while blocking risky actions.



Impact & future scope

- Lower fraud losses through proactive, continuous detection
- Better customer experience via risk-based, minimal friction
- Stronger regulatory posture with auditable policy decisions

Future scope

1. Deepfake detection for biometric and video verification
2. Graph-based fraud analysis to surface coordinated rings
3. Federated learning to share models across institutions while preserving privacy



**"Building
continuous
trust in digital
banking"**